Determining Minimum Sample Size for Estimating a Population Mean

Statistics Worksheet · Grade 11–12 / Introductory College Statistics

Name: ________________________________ Date: __________________

Learning Objectives

- Apply the sample size formula $n \geq (z^* \cdot \sigma / E)^2$ to find the minimum sample size needed for a given confidence level and margin of error.
- Identify and interpret the components of the margin-of-error formula: critical value (z^*), population standard deviation (σ), and maximum error of estimate (E).
- Explain how increasing sample size reduces variability and improves the accuracy of a confidence interval estimate.

Problems

1. A researcher wants to estimate a population mean with a maximum error of 3, a population standard deviation of 12, and a 95% confidence level ($z^* = 1.96$). Use the sample size formula to find the minimum sample size needed. Round up to the nearest whole number.

$$n \geq \left(\frac{z^* \cdot \sigma}{E}\right)^2$$

2. A survey requires a maximum error of 5 units. The population standard deviation is 20 and the confidence level is 90% ($z^* = 1.645$). What is the minimum sample size required?

$$n \geq \left(\frac{z^* \cdot \sigma}{E}\right)^2$$

3. Researchers want to estimate the average weight of adult cats. Previous studies show a population standard deviation of 1.5 pounds. They want their estimate to be within 0.5 pounds at a 95% confidence level ($z^* = 1.96$). What is the minimum number of cats needed?

$$n \geq \left(\frac{1.96 \times 1.5}{0.5}\right)^2$$

4. A study uses a sample of 100 people and achieves a margin of error of 2.45 with $z^* = 1.96$. What was the population standard deviation used in this calculation? Solve for sigma.

$E = z^* \cdot \frac{\sigma}{\sqrt{n}}$

5. A health study estimates the mean blood pressure of adults. The population standard deviation is 8 mmHg. A researcher requires the estimate to be within 2 mmHg. Compare the minimum sample sizes needed at 90% confidence (z* = 1.645) and at 99% confidence (z* = 2.576). How many more participants are needed at the higher confidence level?

$n \geq \left(\frac{z^* \cdot \sigma}{E}\right)^2$

6. A quality control engineer wants to estimate the mean diameter of steel rods produced at a factory. The population standard deviation is 0.04 cm and the desired maximum error is 0.01 cm. Find the minimum sample size at a 99% confidence level (z* = 2.576).

$n \geq \left(\frac{2.576 \times 0.04}{0.01}\right)^2$

7. A researcher currently plans to use a sample size of 50 to estimate a population mean, but their advisor recommends reducing the maximum error by half (from 4 to 2). The population standard deviation is 10 and z* = 1.96. What is the new minimum sample size? By what factor does the sample size change?

$n \geq \left(\frac{z^* \cdot \sigma}{E}\right)^2$

8. A nutritionist wants to estimate the mean daily caloric intake of teenagers. A pilot study shows σ = 250 calories. The researcher wants to be 95% confident (z* = 1.96) with a maximum error of 30 calories. A colleague suggests using a maximum error of 50 calories instead. How much smaller would the sample size be with the larger margin of error? Show both calculations.

$n \geq \left(\frac{z^* \cdot \sigma}{E}\right)^2$

9. A zoologist wants to estimate the mean lifespan of a species of tropical fish. Based on prior research, the population standard deviation is 3.2 years. The zoologist wants a margin of error no greater than 0.6 years and a confidence level of 99% (z* = 2.576). Determine the minimum sample size. Then state what happens to the sampling distribution as this sample size is used — in terms of shape and variability.

$n \geq \left(\frac{2.576 \times 3.2}{0.6}\right)^2$

10. A medical research team is designing a study to estimate the mean recovery time (in days) for patients undergoing a new knee surgery. From historical records, the population standard deviation is 6.5 days. The team is debating between three designs: (A) 95% confidence with E = 1 day, (B) 99% confidence with E = 1 day, and (C) 99% confidence with E = 1.5 days. Using z* = 1.96 for 95% and z* = 2.576 for 99%, compute the minimum sample size for each design. Rank them from smallest to largest and explain which design offers the best balance between precision and feasibility.

$$n \geq \left(\frac{z^* \cdot \sigma}{E}\right)^2$$

Determining Minimum Sample Size for Estimating a Population Mean — Answer Key

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Answer Key

1. Answer: $n = 62$

- Identify the values: $z^* = 1.96$, $\sigma = 12$, $E = 3$.
- Substitute into the formula: $n \geq (1.96 \times 12 / 3)^2$
- Compute the numerator: $1.96 \times 12 = 23.52$
- Divide: $23.52 / 3 = 7.84$
- Square: $7.84^2 = 61.47$
- Round up to the nearest whole number: $n = 62$

2. Answer: $n = 44$

- Identify the values: $z^* = 1.645$, $\sigma = 20$, $E = 5$.
- Substitute: $n \geq (1.645 \times 20 / 5)^2$
- Compute: $1.645 \times 20 = 32.9$
- Divide: $32.9 / 5 = 6.58$
- Square: $6.58^2 = 43.30$
- Round up: $n = 44$

3. Answer: $n = 35$

- Identify the values: $z^* = 1.96$, $\sigma = 1.5$, $E = 0.5$.
- Compute numerator: $1.96 \times 1.5 = 2.94$
- Divide: $2.94 / 0.5 = 5.88$
- Square: $5.88^2 = 34.57$
- Round up: $n = 35$

4. Answer: $\sigma = 12.5$

- Start with the margin of error formula: $E = z^* \cdot \sigma / \sqrt{n}$
- Substitute known values: $2.45 = 1.96 \cdot \sigma / \sqrt{100}$
- Simplify $\sqrt{100} = 10$: $2.45 = 1.96 \cdot \sigma / 10$
- Multiply both sides by 10: $24.5 = 1.96 \cdot \sigma$
- Divide both sides by 1.96: $\sigma = 24.5 / 1.96 = 12.5$

5. Answer: $n(90\%) = 44$, $n(99\%) = 107$; 63 more participants needed

- For 90%: $n \geq (1.645 \times 8 / 2)^2 = (6.58)^2 = 43.30 \rightarrow n = 44$
- For 99%: $n \geq (2.576 \times 8 / 2)^2 = (10.304)^2 = 106.17 \rightarrow n = 107$
- Difference: $107 - 44 = 63$ more participants are needed at the 99% confidence level.
- This illustrates that higher confidence levels require larger sample sizes.

6. Answer: $n = 107$

- Identify: $z^* = 2.576$, $\sigma = 0.04$, $E = 0.01$
- Compute numerator: $2.576 \times 0.04 = 0.10304$
- Divide: $0.10304 / 0.01 = 10.304$
- Square: $10.304^2 = 106.17$
- Round up: $n = 107$

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7. Answer: Original n = 25, New n = 97; sample size increases by a factor of 4

- Original: $n \geq (1.96 \times 10 / 4)^2 = (4.9)^2 = 24.01 \rightarrow n = 25$
- New ($E = 2$): $n \geq (1.96 \times 10 / 2)^2 = (9.8)^2 = 96.04 \rightarrow n = 97$
- Ratio: $97 / 25 \approx 4$
- When the maximum error is halved, the required sample size quadruples. This is because n depends on $(1/E)^2$, so halving E multiplies n by 4.

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8. Answer: n(E=30) = 267, n(E=50) = 97; difference = 170 fewer participants

- For $E = 30$: $n \geq (1.96 \times 250 / 30)^2 = (16.333...)^2 = 266.78 \rightarrow n = 267$
- For $E = 50$: $n \geq (1.96 \times 250 / 50)^2 = (9.8)^2 = 96.04 \rightarrow n = 97$
- Difference: $267 - 97 = 170$ fewer participants when $E = 50$.
- A larger allowable error dramatically reduces the required sample size.

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9. Answer: n = 189; the sampling distribution approaches normality and variability decreases

- Identify: $z^* = 2.576$, $\sigma = 3.2$, $E = 0.6$
- Compute: $2.576 \times 3.2 = 8.2432$
- Divide: $8.2432 / 0.6 = 13.7387$
- Square: $13.7387^2 = 188.75$
- Round up: $n = 189$
- Interpretation: As sample size increases toward 189, by the Central Limit Theorem the sampling distribution of the mean becomes approximately normal and the standard error ($\sigma/\sqrt{n}$) decreases, meaning estimates are more tightly clustered around the true population mean.

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10. Answer: Design A: n = 163, Design B: n = 281, Design C: n = 126; Rank: C < A < B; Design A offers the best balance

- Design A ($z^* = 1.96$, $\sigma = 6.5$, $E = 1$): $n \geq (1.96 \times 6.5 / 1)^2 = (12.74)^2 = 162.31 \rightarrow n = 163$
- Design B ($z^* = 2.576$, $\sigma = 6.5$, $E = 1$): $n \geq (2.576 \times 6.5 / 1)^2 = (16.744)^2 = 280.36 \rightarrow n = 281$
- Design C ($z^* = 2.576$, $\sigma = 6.5$, $E = 1.5$): $n \geq (2.576 \times 6.5 / 1.5)^2 = (11.163)^2 = 124.61 \rightarrow n = 125$; wait — recalculate: $2.576 \times 6.5 = 16.744$; $16.744 / 1.5 = 11.163$; $11.163^2 = 124.61 \rightarrow n = 125$
- Rank from smallest to largest: C (125) < A (163) < B (281)
- Design A balances a high confidence level (95%) with a tight margin of error (1 day) at a feasible sample of 163 patients, making it practical and scientifically rigorous.
- Design B offers the highest confidence but requires 72% more patients than Design A, which may be costly and time-consuming — similar to the monkey cholesterol example where resource constraints matter.
